Musan (Hao) Wu

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Self-Introduction

My name is **Hao Wu**. I prefer to be called by *Hao* (pronounced as "how") or by my English name *Musan* (if you find it ambiguous to say *how are you* to me). I am currently a final-year Ph.D. student at the Institute of Software, University of Chinese Academy of Sciences (UCAS), where I am fortunate to be supervised by Prof. Naijun Zhan. I also completed my undergraduate studies in computer science at the same university.

Research Interests and Selected Publications

My research interests primarily focus on two topics:

- **Formal Verification and Synthesis.** I study how to combine symbolic computation and numerical optimization in formal verification and synthesis, with the goal to advance extensible and scalable formal reasoning methodologies. I investigate a diverse range of infinite-state systems exhibiting various computational behaviors, including discrete-time programs, continuous-time and hybrid systems, control systems, stochastic systems, and communication protocols. The representative works include:
  - [TOPLAS 2025] employs robust optimization techniques to approximate the set of all possible program invariants subject to a certain template.
  - [Festschrift 2025] proposes some techniques to improve the difference-of-convex method for solving non-convex barrier certificate constraints (as bilinear matrix inequalities).
  - [FM 2024] uses polynomial homogenization to extend Sum-of-Squares-based template synthesis to unbounded domains.
- **Complexity/Computability Issues.** I have long been enthusiastic about topics in theoretical computer science, spanning logic, automata, and algorithms, with the aim of deepening our understanding of the complexity and computability issues. Recently I focus on reasoning about nonlinear arithmetic:
  - [(Under Review) 2025] Fourier Motzkin elimination and Cylindrical Algebraic Decomposition can be made single exponential for problems with treewidth sparsity.
  - [FM 2024] gives a complete method based on semidefinite programming to compute the Craig interpolation of two polynomial formulas.
  - [Acta 2024] proves a doubly exponential bound for deciding quantifier-free Presburger arithmetic formulas with 2^x functions, and show how to apply it in string constraint solving.

Education

- **Institute of Software, University of Chinese Academy of Sciences (UCAS)** — *Sep. 2020 - June. 2026 (planned)*
  Ph.D. Candidate in Computer Science at St. Key Lab. Comput. Sci. — Beijing, China
  - Advisor: Prof. Dr. Naijun Zhan
  - First Class Scholarship. Outstanding Student.
- **University of Chinese Academy of Sciences (UCAS)** — *Sep. 2016 - June. 2020*
  B.Sc. (with honor) in Computer Science. (Note:UCAS began to recruit undergraduate students in 2014) — Beijing, China
  - Advisor: Prof. Dr. Xiaoming Sun
  - GPA: 3.86/4. Distinguished B.Sc. Thesis Award.

Short-term academic visit:

- **The Hong Kong University of Science and Technology (HKUST)** — *Mar. 2024 - May. 2024*
  PG Visiting Internship Students (Research) — Hong Kong, China
  - Advisor: Prof. Dr. Amir Goharshady (now at Oxford)

PUBLICATIONS

Due to the doctoral graduation requirements in China, the author order (primarily the *first* author) is determined *by contribution*, and we use this order as the default. Names marked with an asterisk (∗) denote *equal contribution*, and names marked with a dagger (†) indicate *alphabetical order*. Click the titles to get pdf files from my webpage.

In submission.

1. **Hao Wu**, Jiyu Zhu, Amir Goharshady, Jie An, Bican Xia, and Naijun Zhan. **Quantifier Elimination Meets Treewidth**. Submitted to TACAS, 2025.
2. Yihao Yin, **Hao Wu**, Wan Liu, Shuling Wang, Xiong Xu, Wang Lin, Fanjiang Xu, and Naijun Zhan. **Formal Design of Safety-Critical Systems with MARS**. Submitted to Journal of Systems Architecture, 2025.

Published.

1. **Hao Wu**, Qiuye Wang, Bai Xue, Naijun Zhan, Lihong Zhi, and Zhi-Hong Yang. **Synthesizing Invariants for Polynomial Programs by Semidefinite Programming**. ACM Transactions on Programming Languages and Systems, TOPLAS 2025.
2. Xiong Xu, Jean-Pierre Talpin, Shuling Wang, **Hao Wu**, Bohua Zhan, Xinxin Liu, and Naijun Zhan. **HpC: A Calculus for Hybrid and Mobile Systems**. In the ACM Conference on Object-Oriented Programming, Systems, Languages, and Applications, OOPSLA 2025.
3. Guanhua Lin, **Hao Wu**†, and Naijun Zhan. **Barrier Certificate Synthesis via Interpolation and Difference-of-Convex Programming**. In the Proc. of Colloquium on Design and Verification of Cyber-Physical Systems: From Theory to Applications, Festschrift of Prof. Martin Fränzle's 60th birthday, 2025.
4. **Hao Wu**, Yu-Fang Chen, Zhilin Wu, Bican Xia, and Naijun Zhan. **A decision procedure for string constraints with string/integer conversion and flat regular constraints**. Acta Informatica, 2024.
5. **Hao Wu***, Jie Wang*, Bican Xia, Xiakun Li, Naijun Zhan, Ting Gan. **Nonlinear Craig Interpolant Generation Over Unbounded Domains by Separating Semialgebraic Sets**. In the International Symposium on Formal Methods, FM 2024.
6. **Hao Wu**, Shenghua Feng, Ting Gan, Jie Wang, Bican Xia, and Naijun Zhan. **On Completeness of SDP-Based Barrier Certificate Synthesis over Unbounded Domains**. In the International Symposium on Formal Methods, FM 2024.
7. Yihao Yin, **Hao Wu**, Shuling Wang, Xiong Xu, Fanjiang Xu, and Naijun Zhan. **The Design of Intelligent Temperature Control System of Smart House with MARS**. In the CCF International Symposium on Dependable Software Engineering Theories, Tools and Applications, SETTA 2024.
8. Shuling Wang, Bohua Zhan, Huanhuan Sheng, **Hao Wu**, Shicheng Yi, Lingtai Wang, Xiangyu Jin, Bai Xue, Jinghui Li, Shuang-Qing Xiang, Zhan Xiang, and Bifei Mao. **Survey on Requirements Classification and Formalization of Trustworthy Systems** (in Chinese). Journal of Software, 2022.
9. Xiaohui Bei, Xiaoming Sun, **Hao Wu**†, Jialin Zhang, Zhijie Zhang, and Wei Zi. **Cake Cutting on Graphs: A Discrete and Bounded Proportional Protocol**. In the ACM-SIAM Symposium on Discrete Algorithms, SODA 2020.

ACADEMIC SERVICES

- **Teaching Assistant.**
  - 2023 Spring. Foundations of Theoretical Computer Science, B.Sc., UCAS
- **External Reviewer.**
  - Conferences: TACAS'25, HSCC'25, FM'24, iFM'24, ISSAC'23, CDC'23, TACAS'23, ChinaSoft'23, ICTAC'22
  - Journals: TAC, Automatica, Journal of Symbolic Computation (JSC), TCAD.
- **Projects.**
  - Participant in NSFC Key International Joint Research Program (China, France): Research on Design Techniques for Safety-Critical Cyber-Physical Systems in the Context of Mobile Computing
  - Participant in National Key R&D Program of China under grant No. 2022YFA1005101
  - Participant in Natural Science Foundation of China under grant No. 62192732.

ADDITIONAL INFORMATION

Languages: English (working language), Mandarin Chinese (native)

REFERENCES

1. **Prof. Dr. Naijun Zhan**
   Boya Distinguished Professor, School of Computer Science
   Peking University (PKU)
   Email: znj@ios.ac.cn
   Relationship: Advisor

2. **Prof. Dr. Amir Goharshady**
   Associate Professor, Department of Computer Science
   University of Oxford
   Email: amir.goharshady@cs.ox.ac.uk
   Relationship: Advisor of HKUST *Internship*

3. **Prof. Dr. Zhilin Wu**
   Full Research Professor, Key Laboratory of Systems Software
   Institute of Software, Chinese Academy of Sciences (CAS)
   Email: wuzl@ios.ac.cn
   Relationship: Collaborator